Rajalakshmi Engineering College

Department of Computer Science and Engineering

LAB RECORD

Library Book Issue Request Form (Advanced) - Advanced Fields

Detail	
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Course / Section	B E CSE
Platform	Zoho Creator
Application	Library Book Issue Management (Advanced Edition)
Assignment	Assignment 06 - Advanced Fields

1. Aim

To extend the Book_Issue_Request_Form in Zoho Creator by adding all seventeen Advanced Fields, apply the required validations and configurations, and extend the Book_Issue_Request_Report with the computed fine, issue request ID and QR code columns, so that catalog verification, multi-book requests, fine computation and scannable issue-request references are supported.

2. Procedure

2.1 Extending the application and form

1. Opened the existing application Library Book Issue Management and renamed it to Library Book Issue Management (Advanced Edition).
2. Created a master form named Book_Catalog holding the book title, author, ISBN and shelf location, and added a few sample books.
3. Opened Book_Issue_Request_Form in the Form Builder and added four new sections: E - Catalog Verification & Reference, F - Multi-Book Request & Media Details, G - Fine & Delivery Preference, H - Declaration & System-Generated Values.

2.2 Adding the seventeen Advanced Fields

Section	Field name	Field type	Setting
E	Book_Title_Lookup	Lookup	Connected to master Book Catalog; auto-fills author, ISBN, shelf location
E	Book_Cover_Image	Image	JPG and PNG only; max 2 MB
E	Ebook_Reference_URL	URL	Link to online e-book version
E	Book_Replacement_Price	Decimal	Replacement price of the book
F	Additional_Books_Requested	SubForm	Min 0, max 4 rows
F	Faculty_Recommendation_Upload	File Upload	PDF, JPG, PNG; max 5 MB
F	Book_Review_Video	Video	Optional; max 45 seconds and 50 MB

Section	Field name	Field type	Setting
F	Special_Request_Voice_Note	Audio	Optional voice note
G	Requires_Home_Delivery	Decision Box	Mandatory
G	Fine_Amount_Due	Currency	INR (₹); non-negative values only
G	Late_Fee_Waiver_Percent	Percent	Range 0 to 100; editable by library staff only
G	Total_Fine_Payable	Formula	Read-only; Fine_Amount_Due - (Fine_Amount_Due x Late_Fee_Waiver_Percent)
H	Library_Rules_Display	Rich Text	Read-only reference block
H	Member_Digital_Signature	Signature	Mandatory
H	Request_Submitted_On	Date-Time	Auto-captured on submission
H	Issue_Request_ID	Auto Number	Prefix LIB-ISS-, current year, 6-digit sequence
H	Book_Issue_QR_Code	QR/Barcode	Auto-generated from Issue_Request_ID

2.3 Configuring validations and rules

1. Connected Book_Title_Lookup to the Book_Catalog form so that members search and select the title instead of typing it, and set the author, ISBN and shelf location to auto-fill.
2. Configured the Additional_Books_Requested SubForm with a minimum of 0 and a maximum of 4 rows, with columns for title, author and category.
3. Restricted Faculty_Recommendation_Upload to PDF, JPG and PNG files with a maximum size of 5 MB.
4. Restricted Book_Cover_Image to JPG and PNG files with a maximum size of 2 MB.
5. Restricted Book_Review_Video to a maximum duration of 45 seconds and a maximum size of 50 MB.
6. Limited Late_Fee_Waiver_Percent to the range 0 to 100 and made it editable only by library staff.
7. Set Fine_Amount_Due to accept non-negative values only, formatted in INR (₹).
8. Set Total_Fine_Payable as a read-only formula field that recalculates whenever Fine_Amount_Due or Late_Fee_Waiver_Percent changes.
9. Configured Issue_Request_ID with the prefix LIB-ISS-, the current year and a 6-digit auto-incrementing sequence.
10. Configured Book_Issue_QR_Code to generate automatically from Issue_Request_ID after submission.
11. Marked Member_Digital_Signature and Requires_Home_Delivery as mandatory so that the form cannot be submitted without them.

2.4 Report

1. Edited the Book_Issue_Request_Report to add the Total_Fine_Payable, Issue_Request_ID and Book_Issue_QR_Code columns.
2. Opened the form in Preview mode, filled it in, and submitted three sample book issue requests.

3. Output

(a) Form Builder showing all 12 Advanced Fields

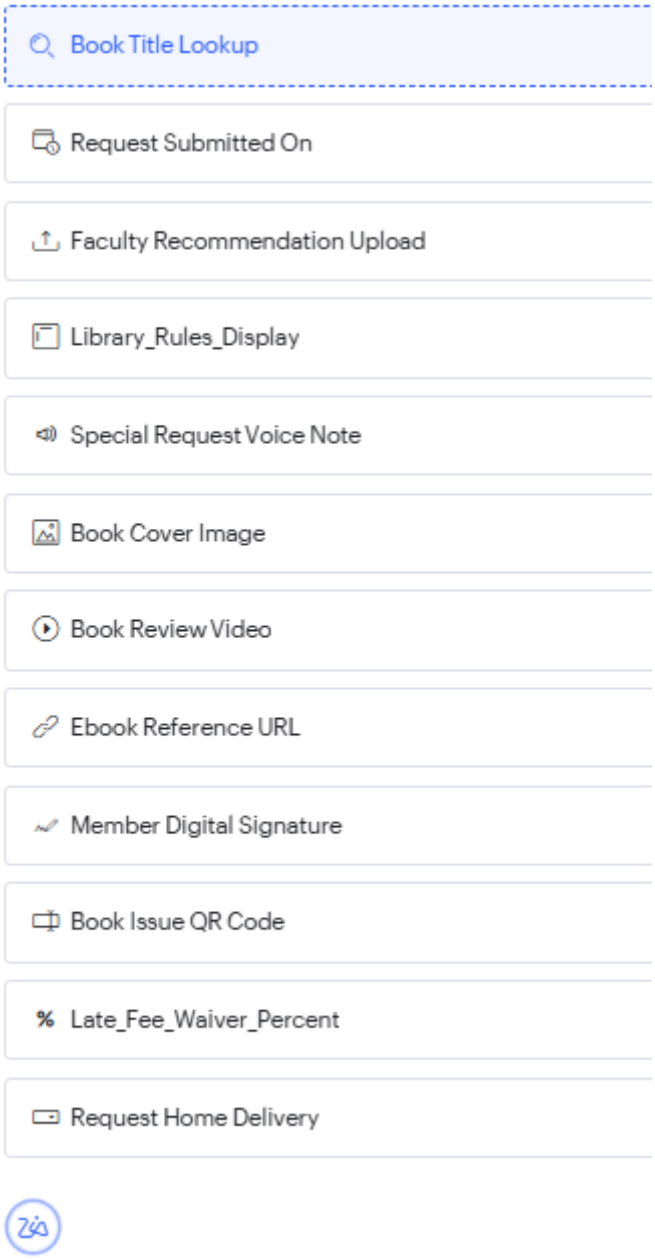


Figure 1: Form Builder showing Advanced Fields

(b) Live preview of the form

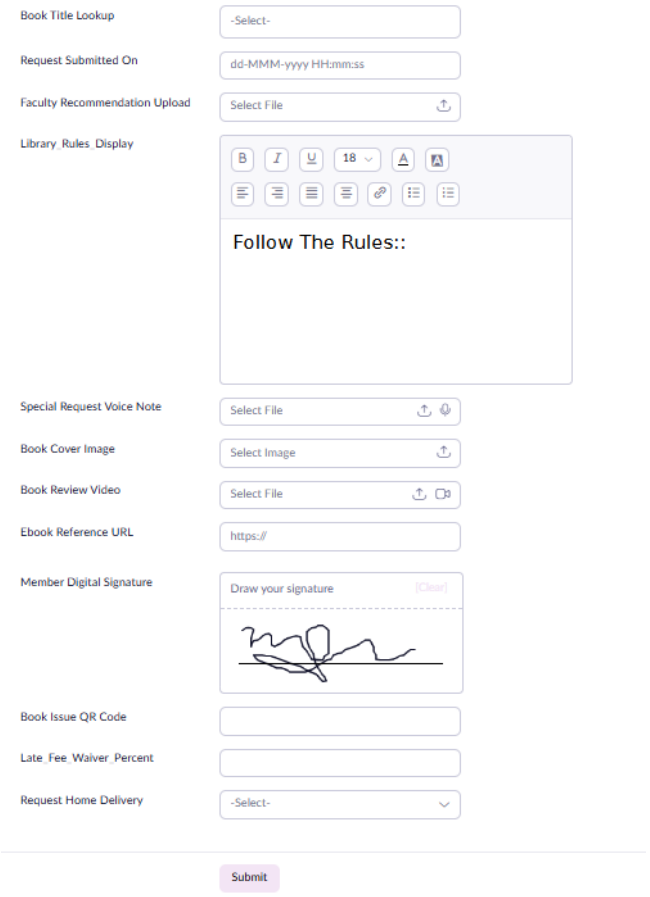


Figure 2: Live preview of Book_Issue_Request_Form

Result:

The Book_Issue_Request_Form was extended in Zoho Creator with all seventeen Advanced Fields arranged in four new sections. The required validations and configurations were applied, and the Book_Issue_Request_Report displays the submitted sample records with the computed fine, issue request ID and QR code.